

❏ 1. Accreditation Standards in Cyber Forensics

❏ What is Accreditation?

Accreditation is a **formal recognition** by an authorized body that a **forensics laboratory** or **investigator** is competent to carry out specific tasks **according to international standards**.

❏ Why Accreditation Matters:

- Ensures **credibility** of forensic evidence in court
- Follows **chain of custody** and **data handling** rules
- Verifies that tools and methods meet **quality benchmarks**
- Guarantees **repeatability** and **reproducibility** of findings
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Enables cross-border cooperation in investigations

☒ **Common Accreditation Standards:**

Standard	Description
ISO/IEC 17025	For testing/calibration laboratories – ensures technical competence and quality system
ISO/IEC 27037	Guidelines for identification, collection, acquisition, and preservation of digital evidence
ISO/IEC 27042	Guidelines for analysis and interpretation of digital evidence
NIST SP 800-101	Guidelines on Mobile Device Forensics
SWGDE & SWGIT	Best practices for digital and multimedia evidence (U.S. based)
FISMA/NIST 800-53	Security standards for federal information systems

☒ **Tool Accreditation:**

Forensic tools (like EnCase, FTK, Autopsy) should be **validated and verified**:

- Tools must produce **repeatable** and **verifiable** results
- Labs often maintain a **tool validation repository**

☒ 2. Performing a Cyber Forensics Investigation

A typical cyber forensics investigation involves multiple structured phases:

☒ Investigation Lifecycle:

Phase	Description
1. Preparation	Define scope, obtain legal authorization, prepare tools

2. Identification	Identify affected systems, sources of evidence (logs, devices, memory)
3. Preservation	Use write blockers, make forensic images, maintain chain of custody
4. Collection	Collect logs, memory, drives, network packets, etc.
5. Examination	Use tools to search for deleted files, malware, timestamps
6. Analysis	Reconstruct timeline, detect attacker actions, recover passwords
7. Documentation	Record every action taken, evidence logs, hashes, findings
8. Reporting	Provide detailed, formal report; include screenshots, artifacts, tools used
9. Presentation (in court)	Explain findings clearly and legally admissibly

☒ Common Tools in Use:

- Disk & memory imaging: FTK Imager, dd, Magnet RAM Capture
- Analysis: Autopsy, EnCase, X-Ways, Volatility
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Network forensics: Wireshark, TCPDump

- **Log analysis:** Splunk, ELK Stack

☒ 3. Privacy and Cyber Forensics

☒ Key Privacy Concerns:

- **Right to privacy vs. need for investigation**
- **Handling PII (Personally Identifiable Information)**
- **Legal compliance (GDPR, HIPAA, IT Act, etc.)**
- **Ensuring data minimization and consent-based access**

☒ Forensic Analyst's Responsibilities:

- Avoid **unnecessary exploration** of private data
- Maintain **confidentiality** of all findings
- Never exceed the **scope of authorization**
- Anonymize or redact non-relevant PII in reports
- Securely store and dispose of sensitive evidence

☒ Legal Acts to Know:

- GDPR (EU): Consent, data protection, right to erasure
 - IT Act 2000 (India): Unauthorized access, identity theft
 - HIPAA (U.S.): Protects medical records
 - Privacy Act (U.S.): Controls access to federal data
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☒ Summary:

Area	Importance
Accreditation	Ensures evidence is legally valid and reliable
Investigation Process	Follows 9-phase structured model from preparation to reporting
Privacy	Forensic investigators must balance investigation needs with privacy rights

